L11

29/09/2020

$$\dot{q}(t) = Aq(t) + Bu(t) + L(\hat{y}(t) - y(t))$$
$$\hat{y}(t) = Cq(t) + Du(t)$$
$$\hat{x}(t) = q(t)$$

All the states are not available.

$u = Fx(t)$ ← place eigenvalues of closed loop system

$u = F\hat{x}$ $\quad \hat{x} \to$ estimate

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & BF \\ -LC & A+BF+LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} \rightarrow \begin{bmatrix} \dot{x} - \dot{\hat{x}} \\ \dot{\hat{x}} \end{bmatrix} = \underbrace{\left[\begin{array}{c|c} A+LC & 0 \\ \hline -LC & A+BF \end{array}\right]}\begin{bmatrix} x-\hat{x} \\ \hat{x} \end{bmatrix}$$

estimated

e

Observer

$$\begin{bmatrix} \dot{e} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} \boxed{A+LC} & 0 \\ -LC & \boxed{A+BF} \end{bmatrix} \begin{bmatrix} e \\ \hat{x} \end{bmatrix}$$

Closed loop $u = Fx$

$u = Fx$
$u = F\hat{x}$

Seperation principle

e.g. | Poles: $\{-1, -1, -2\}$ $\quad$ Observer poles: $\{-2, -2, -3\}$

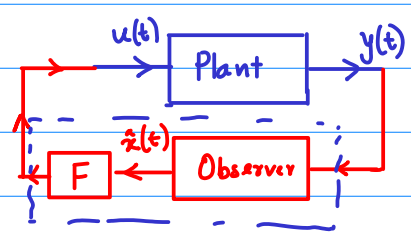
$\sigma(A) = \{1, 1, 2\}$

Observer-based controllers

$$F = \begin{bmatrix} -4 & -4 & 0 \\ 0 & 0 & -4 \end{bmatrix} \qquad L = \begin{bmatrix} -6 & 0 \\ -9 & 0 \\ 0 & -5 \end{bmatrix}$$

$u = F\hat{x}$

Transfer f^n of the controller:



$$\dot{\hat{x}} = -LCx + (A+BF+LC)\hat{x}$$
$$= -L(y - Du) + (A+BF+LC)\hat{x}$$

$y = Cx + Du$
$\Rightarrow Cx = y - Du$
$u = F\hat{x}$

Observer based controller

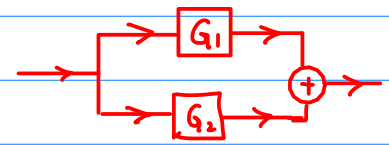
$$\boxed{\begin{aligned} \dot{\hat{x}} &= (A+LC+BF+LDF)\hat{x} - Ly \\ u &= F\hat{x} \end{aligned}}$$

$$K(s) \equiv \begin{bmatrix} (A+LC+BF+LDF) & -L \\ F & 0 \end{bmatrix} \qquad G(s) \equiv \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

e.g. $$K(s) = \begin{bmatrix} (24-60s)/\underline{\underline{s^2-20}} & 0 \\ 0 & -20/\underline{\underline{s-1}} \end{bmatrix}$$ Stabilizing controller

Realization & T.F:

$$G_1 + G_2 = \left[\begin{array}{cc|c} A_1 & 0 & B_1 \\ 0 & A_2 & B_2 \\ \hline C_1 & C_2 & D_1+D_2 \end{array}\right]$$



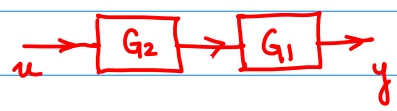
$$G_1 = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \qquad G_2 = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

$$G_1 = \frac{1}{s-1} \qquad G_2 = \frac{s-1}{s+1}$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$$

Series connection:



$G_1 G_2$

$$G_1 G_2 \sim \begin{bmatrix} A_1 & B_1C_2 & B_1D_2 \\ 0 & A_2 & B_2 \\ C_1 & D_1C_2 & D_1D_2 \end{bmatrix} \sim \begin{bmatrix} A_2 & 0 & B_2 \\ B_1C_2 & A_1 & B_1D_2 \\ D_1C_2 & C_1 & D_1D_2 \end{bmatrix}$$

Defination: Dual system: $G(s) = C(sI-A)^{-1}B + D$

$$G(s) \longmapsto G^T(s) = B^*(sI-A^*)^{-1}C^* + D^*$$

e.g. $\frac{1}{s+1} \xrightarrow{\text{Dual}} \frac{1}{s+1}$

$$\begin{bmatrix} 1/s+1 & 1/s \\ 0 & 1/s+1 \end{bmatrix} \xrightarrow{\text{Dual}} \begin{bmatrix} 1/s+1 & 0 \\ 1/s & 1/s+1 \end{bmatrix}$$

Def^n: Conjugate system $\quad G(s) \xrightarrow{\text{conjugate}} G^\sim(s) = -B^*[-sI-A^*]^{-1}C^* + D^*$

$$G^\sim(s) = G^T(-s)$$

e.g. $$\begin{bmatrix} \frac{1}{-s+1} & 0 \\ \frac{1}{-s} & \frac{1}{-s+1} \end{bmatrix}$$

Def^n: $G(s) \in \mathbb{R}^{p \times m}(s)$ ← real rational matrix

$\hat{G}(s) \in \mathbb{R}^{p \times m}(s)$ is called inverse of $G(s)$ if $G(s)\hat{G}(s) = \hat{G}(s)G(s) = I$.

$$G^{-1} \sim \begin{bmatrix} A - BD^{-1}C & -BD^{-1} \\ D^{-1}C & D^{-1} \end{bmatrix}$$